MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 04
Chapter 10: Circles

Time: 45 Minutes
MM: 20

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Case Study based question of 4 marks with internal subparts.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. A circle touches all four sides of a quadrilateral $PQRS$ with $PQ = 11$ cm, $QR = 8$ cm, and $RS = 7$ cm. The length of SP is: [1]

(A) 9 cm (B) 10 cm (C) 12 cm (D) 14 cm

Q2. If the angle between two tangents drawn from a point P to a circle of radius a and centre O is $90°$, then OP is equal to: [1]

(A) a (B) $a\sqrt{2}$ (C) $a/\sqrt{2}$ (D) $2a$

Q3. In two concentric circles of radii 10 cm and 8 cm, the length of a chord of the outer circle which touches the inner circle is: [1]

(A) 6 cm (B) 12 cm (C) 16 cm (D) 18 cm

Q4. Tangents TP and TQ are drawn from external point T to a circle with centre O. If $\angle OPT = 90°$ and $\angle PTQ = 40°$, then $\angle OPQ$ is: [1]

(A) $15°$ (B) $20°$ (C) $25°$ (D) $30°$

Q5. **Assertion (A):** The tangent at any point of a circle is perpendicular to the radius through the point of contact.
Reason (R): The shortest distance of a point from a given straight line is the perpendicular distance. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. From an external point P, two tangents PA and PB are drawn to a circle with centre O. If $\angle AOB = 130°$, find the measure of $\angle APB$. [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. If a, b, c are the sides of a right triangle where c is the hypotenuse, prove that the radius r of the circle which touches the sides of the triangle is $r = \frac{a+b-c}{2}$. [3]

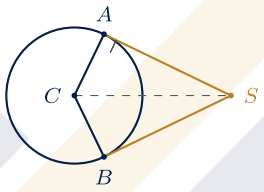
Q8. From an external point P, two tangents PA and PB are drawn to a circle with centre O. If chord AB subtends an angle of $100°$ at the centre, find $\angle APB$ and $\angle PAB$. [3]

Q9. Two concentric circles have radii 13 cm and 8 cm. AB is a diameter of the larger circle and BD is a tangent to the smaller circle touching it at D and meeting the larger circle at E. Find the length of AD. [3]

SECTION D: Case-Based / Applied Competency Question (1 × 4 = 4 Marks)

Q10. Satellite Horizon Line-of-Sight Coverage

A communications satellite orbiting at point S maintains line-of-sight contact with the Earth's horizon at tangent stations A and B. Earth is modeled as a sphere whose cross-section is a circle of radius $R = 6400$ km with centre C. The distance from the satellite to Earth's centre is $SC = 10000$ km.



(i) State why triangle $\triangle CAS$ is a right-angled triangle. [1]

(ii) Calculate the distance SA from the satellite to horizon station A. [1]

(iii) Determine the cosine of angle $\angle ACS$, OR find the distance of the satellite above the Earth's surface. [2]

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